MUHAMMAD ANAS

AI/ML Engineer & Python Developer — Voice & Chat AI Agents, RAG Pipelines & Data Systems

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PROFESSIONAL SUMMARY

AI/ML engineer and Python developer who ships complete, working systems. Built a real-time voice support agent (live at verveassist.live) and a production, Shopify-integrated AI support agent (Gemini embeddings, FAISS vector search, conflict detection, live order lookups) validated by an automated evaluation harness, plus end-to-end applications spanning credit-risk modelling, database design, REST APIs, desktop software, and BI reporting. Comfortable owning a project from data and schema design to a deployed user interface.

TECHNICAL SKILLS

AI / RAG Engineering: Google Gemini API, FAISS vector search, embeddings, LangChain text splitters, prompt design, AssemblyAI (speech-to-text), ElevenLabs (text-to-speech)

Machine Learning: Linear Regression, Decision Trees, Random Forest, SVM, k-NN, Naive Bayes, Clustering, PCA, XGBoost, SHAP, Optuna

Backend & APIs: Flask, Flask-Login, Flask-WTF, Flask-Limiter, Flask-SQLAlchemy, REST API design, Shopify Admin API, Flask-SocketIO

Databases: PostgreSQL, MySQL, SQLite (3NF schema design, joins, indexes, transactions)

Data & Visualization: NumPy, Pandas, Scikit-learn, TensorFlow, Matplotlib, Power BI (DAX, Power Query), Streamlit

Frontend & Desktop: HTML, CSS, JavaScript, Jinja2, Tailwind CSS, CustomTkinter, PyInstaller

Security & Tools: CSRF protection, rate limiting, bcrypt hashing, Git, GitHub, Docker, AWS EC2, Nginx

EXPERIENCE

AI & Machine Learning Intern — Big Brains

Sep – Oct 2026

- Built Loanlytics, an end-to-end credit default risk predictor on 150,000 borrowers (6.7% default rate): XGBoost at 0.864 cross-validated ROC-AUC and 0.870 on a held-out test set
- Benchmarked logistic regression, random forest and XGBoost with 5-fold CV, tuned with Optuna, and set the decision threshold on out-of-fold data to avoid test-set leakage
- Explained predictions with SHAP and shipped a live Streamlit app; completed the program with a 99% average score

AI Intern — Decode Labs (Remote)

August • 2026

- Built a rule-based chatbot handling intent matching and multi-turn conversational logic
- Trained an Iris species classifier using classical supervised learning, evaluating accuracy across multiple algorithms
- Built a tech-stack recommendation tool matching project requirements to suitable technologies

PROJECTS & DELIVERED SYSTEMS

Verve Voice Agent — Real-Time Voice Support Platform

2026 • Lahore

Flask-SocketIO, AssemblyAI, Gemini, ElevenLabs, FAISS, AWS EC2

- Built a real-time voice support agent for the AssemblyAI Voice Agent Hackathon that answers from a brand's own policy documents and hands off to a human when sources conflict or a message is sensitive
- Chained speech-to-text, FAISS retrieval, conflict check, routing and spoken reply into one pipeline; deployed live on AWS EC2 at verveassist.live

AI Support Agent — Shopify-Integrated RAG Platform

2026 • Lahore

Flask, Gemini API, FAISS, Shopify Admin API, Nginx

- Built a Shopify-integrated AI agent answering policy Q&A via FAISS retrieval, flagging conflicts
- Built an evaluation harness (Q&A, conflict detection, routing) achieving **98% accuracy across 59 tests**
- Integrated Shopify Admin API for real-time order lookups; rate-limited chat to **10 requests/minute**
- Deployed via systemd behind Nginx with Let's Encrypt SSL; secured admin login with session-based auth

Hotel & Reservation Management System — Full-Stack Web App

2026 • Lahore

Python (Flask REST API), PostgreSQL, HTML/CSS/JavaScript, Docker, AWS EC2

- Designed a 3NF PostgreSQL schema (10 tables) with FKs and CHECK constraints across Lahore & Islamabad branches
- Wrapped reservation and payment creation in one atomic transaction; built a real-time occupancy/revenue dashboard
- Migrated MySQL to PostgreSQL, containerized with Docker, and deployed on AWS EC2 behind Nginx with SSL

StockFlow Pro — Inventory, Sales & Warehouse Management System *2026 • Lahore*

Python, CustomTkinter, MySQL

- Shipped a barcode-level warehouse system as a standalone Windows .exe requiring no Python installation
- Implemented a per-carton audit trail with embedded Matplotlib reporting and one-click PDF export via ReportLab

SellerDocs AI — RAG Document Assistant for E-Commerce Sellers *2026 • Lahore*

Flask, Gemini API, FAISS, SQLite, Docker

- Built a multi-tenant RAG assistant with per-topic knowledge bases and cross-document conflict detection
- Built an evaluation module tracking retrieval accuracy against test questions with expected sources

EDUCATION

BS Data Science — University of Management & Technology (UMT), Lahore *2024 – Present*

Relevant coursework: Machine Learning, Data Analysis (Power BI), Statistics, Database Management, Python Programming

CERTIFICATIONS

- AI & Machine Learning Internship — Big Brains • Oct 2026 • 99% average score
- Artificial Intelligence using Python — DigiSkills.pk • Jul 2026
- Data Analytics and Business Intelligence — DigiSkills.pk • Mar 2026
- Data Science & Analytics — HP LIFE • Aug 2025

Languages: Urdu (Native) • English (Fluent)